

Full Stack Mobile Application

Design under Web and Mobile Technologies module

20-04-2026

1. Project Overview & System Flow

The project is a comprehensive **Personalized E-Commerce Mobile Application** designed to transform the traditional online shopping experience. Beyond standard retail operations, this system specializes in tailored gift items (such as Sprout Pencils and Magic Jars) and bespoke memory products (such as Custom Anniversary Memory Cards).

The operational workflow of the application flows seamlessly through three primary phases:

A. Authentication & Security Layer

- **Access Control:** While public users can browse products, any interactive operation such as customizing a gift, managing a cart, or checking out—requires secure user authentication.
- **Cryptographic Protection:** During user registration, passwords undergo one-way cryptographic hashing using **bcrypt** before database storage.
- **Session Management:** Upon successful login, the backend issues a **JSON Web Token (JWT)**. This token is securely stored on the mobile client and transmitted in HTTP headers to access protected private routes (e.g., placing orders or updating customization profiles).

B. Core Customer Journey

1. **Product Discovery:** Customers explore the application via the "Shop Grid" (**Member 1: Product & Inventory Manager**), which dynamically fetches available gift items from the cloud database.
2. **Bespoke Customization:** When choosing a customizable item, the customer is directed to the "Build Your Gift" form (**Member 3: Personalized Customization Specialist**), where they select typography, themes, and input custom textual content.
3. **Media Integration:** To add a personal touch (e.g., photos for memory cards), the application activates a native camera/gallery image picker, routed through **Multer** processing to handle dynamic image uploads and metadata tracking (**Member 6: Media Storage & Deployment Engineer**).
4. **Discount Evaluation:** Prior to checkout, customers can enter active coupon codes (e.g., "EGO2026") within the cart interface. The system processes the request through the

validation logic engine (**Member 4: Promotions & Discount Engine**) to instantly calculate deductions.

5. **Order Finalization:** The validated transaction proceeds to the "Checkout" pipeline (**Member 2: Order & Transaction Lead**), generating an immutable purchase log with distinct status tracking.
6. **Social Proof & Feedback:** Upon delivery, customers utilize the interactive star-rating and critique component (**Member 5: Customer Feedback & Rating Moderator**) embedded within the product views.

C. Admin & Back-Office Pipeline

- **Inventory Control:** Admins utilize a dedicated, secure dashboard to dynamically modify pricing structures and track real-time stock levels.
- **Order Fulfilment:** Administrators manage the order pipeline by transitioning transaction states sequentially from **Pending** $\rightarrow$ **Processing** $\rightarrow$ **Delivered**.
- **Campaign Management:** Authorized personnel can inject new promo codes, adjust promotional expiration dates, and discard obsolete campaigns.

2. Technology Stack Architecture

To achieve low latency, cross-platform stability, and effortless scaling, the system leverages a production-grade modern JavaScript/TypeScript ecosystem:

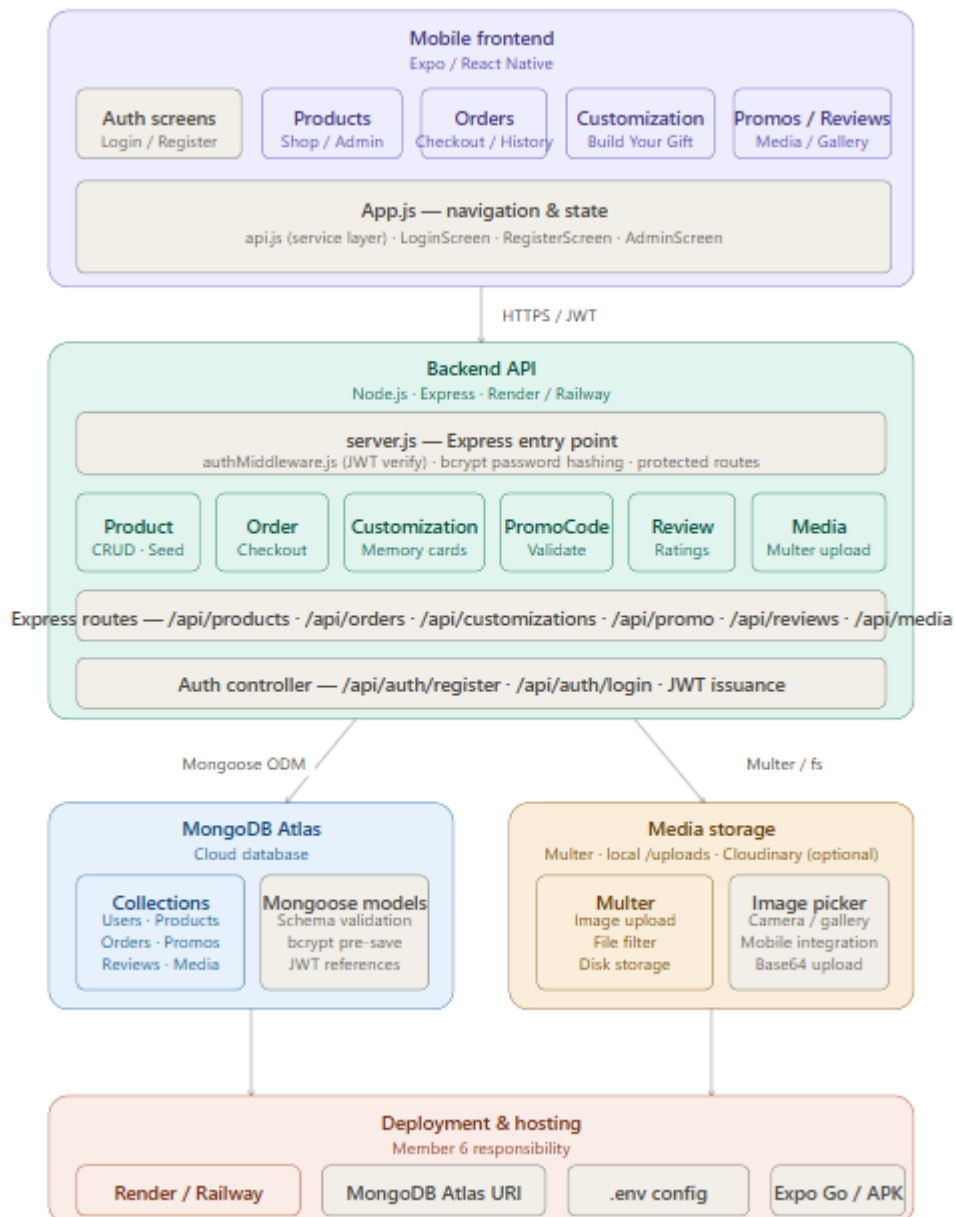
- **Frontend Framework: Expo & React Native**
 - *Justification:* Utilizing a unified JavaScript codebase allowed the team to cross-compile the application natively into high-performance binaries for both iOS and Android platforms, cutting development overhead in half while maintaining a smooth native UI feel.
- **Backend Server: Node.js & Express.js**
 - *Justification:* Express.js offers an asynchronous, event-driven architecture well-suited for I/O-intensive web applications. It serves as a highly scalable RESTful API broker that efficiently processes concurrent user operations and database transactions.
- **Cloud Database: MongoDB Atlas**
 - *Justification:* Since customized gifts involve semi-structured, highly dynamic data schemas (varying fonts, flexible review structures, diverse image arrays, and

unique text strings), a schema-less NoSQL document store like MongoDB was preferred over restrictive relational databases.

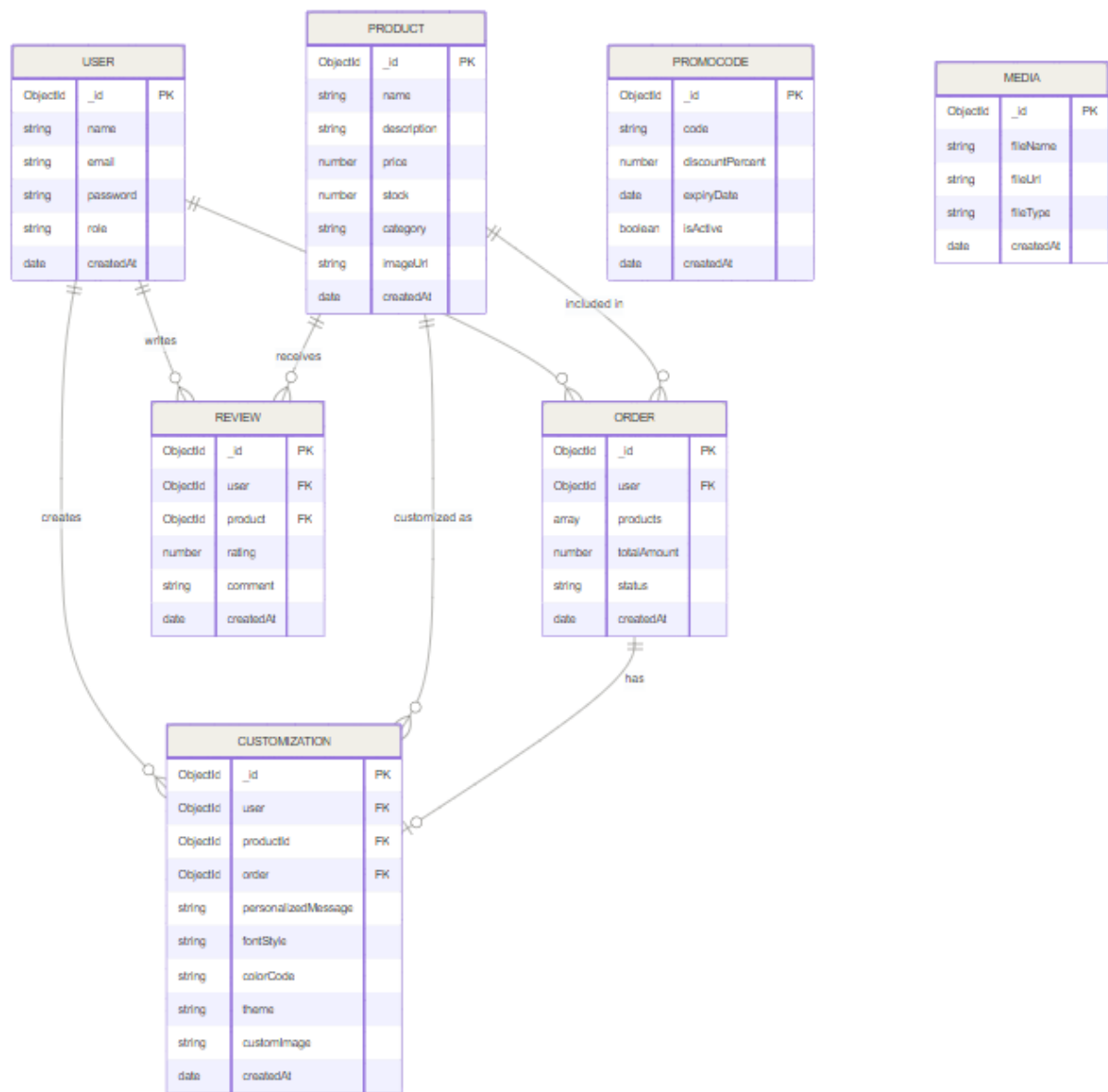
- **Deployment Infrastructure: Vercel**

- *Justification:* Deploying the Node.js API to Vercel provided a serverless cloud infrastructure ensuring maximum uptime, automated SSL encryption out-of-the-box, and optimized content delivery speeds for api routing.

3. System Architecture Diagram



4. Data Schema Diagram



5. Deployment Details

Backend URL: <https://ego-creation-project.onrender.com>

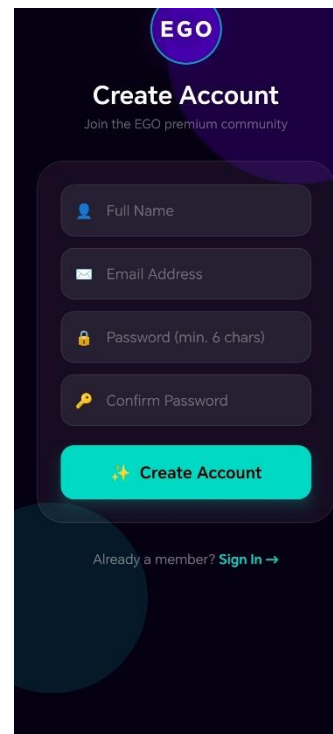
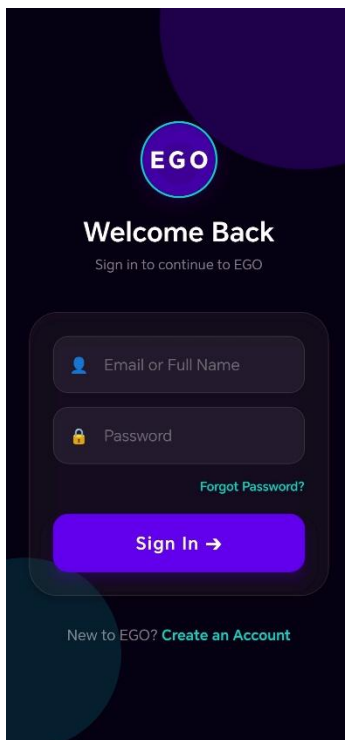
GitHub Repository: https://github.com/IT24101896/Ego_Creation_Project.git

Frontend:

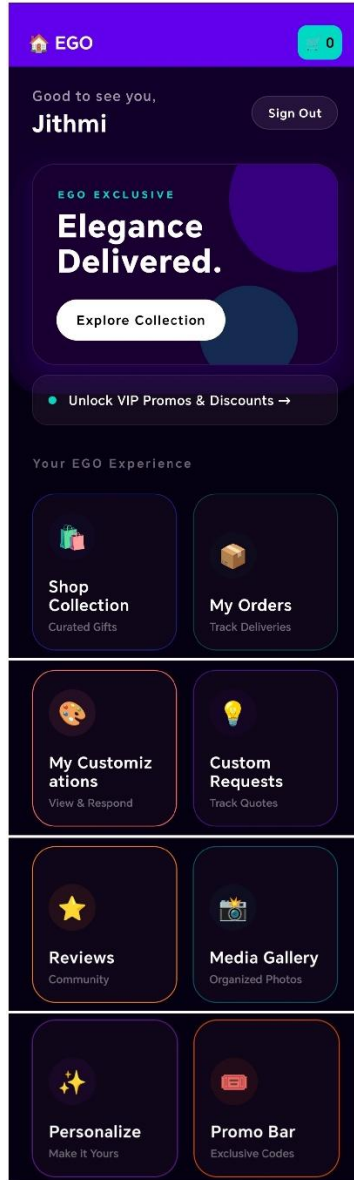


6.App Development Proof (Screen Shots)

I. Logging Dashboard



II. User Dashboard



III. Admin Dashboard



Admin Responsible pages

← Back

Admin Center

ProductsOrdersCus

+ Add New ItemCurrent Items (9)

Add to Inventory

Product Name *

Price (\$) *Stock *

Description

Tap to Upload Image

Add Product

← Back

Admin Center

ProductsOrdersCus

+ Add New ItemCurrent Items (9)

Jar

\$500.00 · Stock: 55

Flowers

\$1000.00 · Stock: 100

Cards

\$100.00 · Stock: 20

Chocolate

\$150.00 · Stock: 500

Toys

\$50.00 · Stock: 100

Statues

\$5000.00 · Stock: 10

← Back

Admin Center

ProductsOrdersCus

Order #7997FC

\$45.00

User: Jithmi

Shipped

Mark Delivered

Order #79979F

\$540.00

User: Jithmi

Delivered

Order #B3FEEE

\$500.00

User: kasun

Delivered

Order #B3FEEC

\$250.00

User: kasun

Delivered

← Back

Admin Center

ProductsOrdersCus

Pending Customizations

Review and approve customer customizations. Adjust prices as needed.

U

Unknown User

No email

02/05/2024

PENDING

Customization Details:

Message: "I love you"

Font: Elegant

Color: Gold

Product:

Flowers

Suggested Price:

0.00

Approve

Reject

← Back

Admin Center

nsCustom RequestsPending

Custom Item Requests

Review customer requests for custom items and provide quotes.

Create Custom Request for Customer

J

Jithmi

jithmi@gmail.com

23/05/2024

In Progress

Requested Item:

Key tag

Category: Stationary

Budget: \$0 - \$25

Urgency: Normal

I want it in beauty

Personalization:

Message: "I love you"

← Back

Admin Center

viewsPromosUsers

Create Promo Code

Create instant promo codes that users can apply immediately

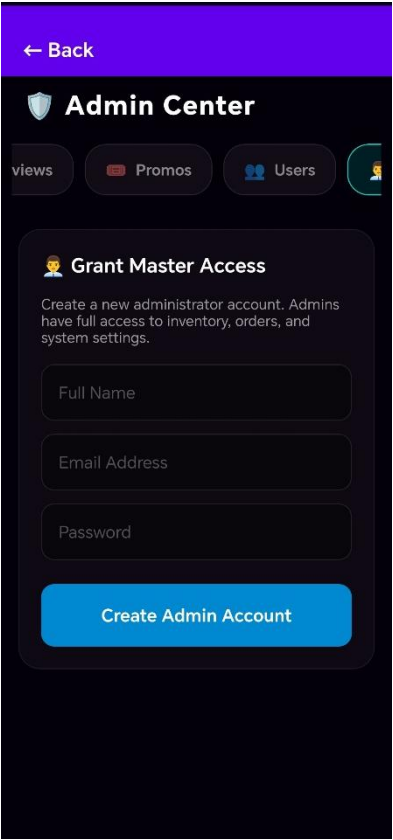
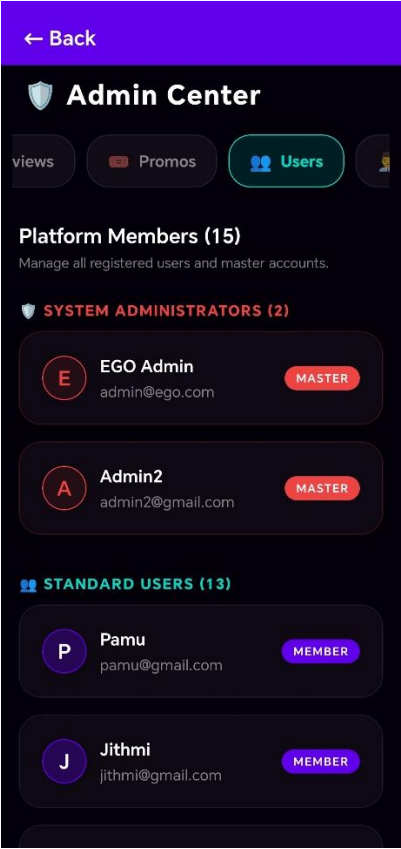
Promo Code (e.g. SUMMER20)

Discount %Min Order \$

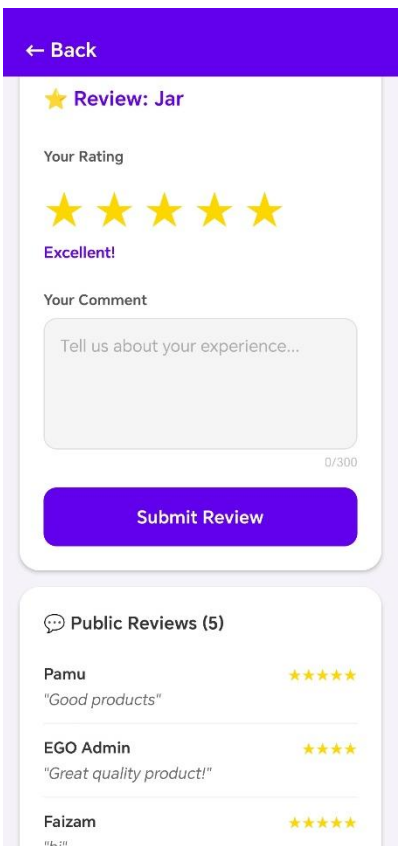
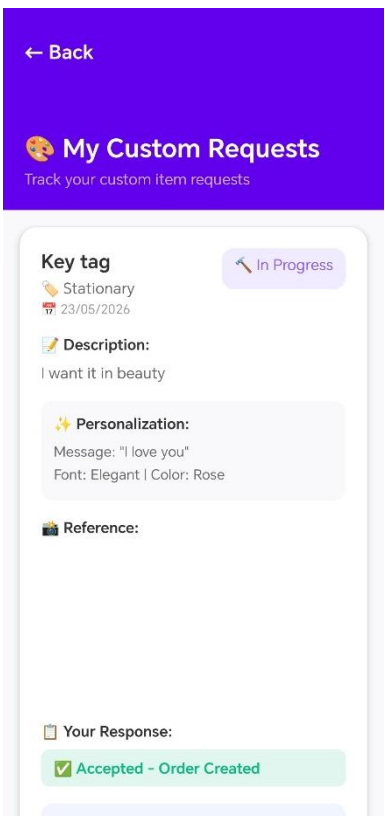
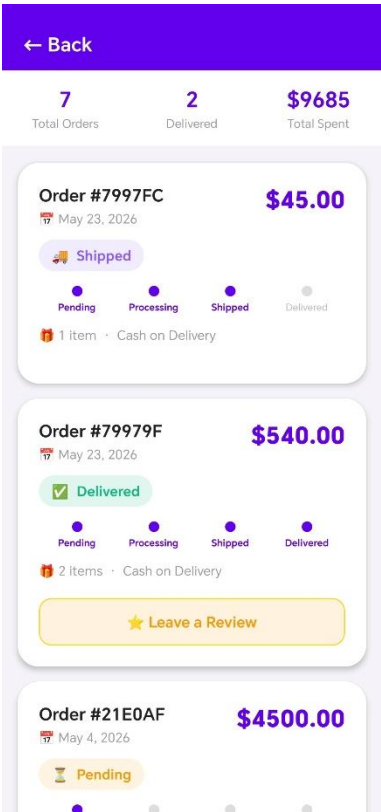
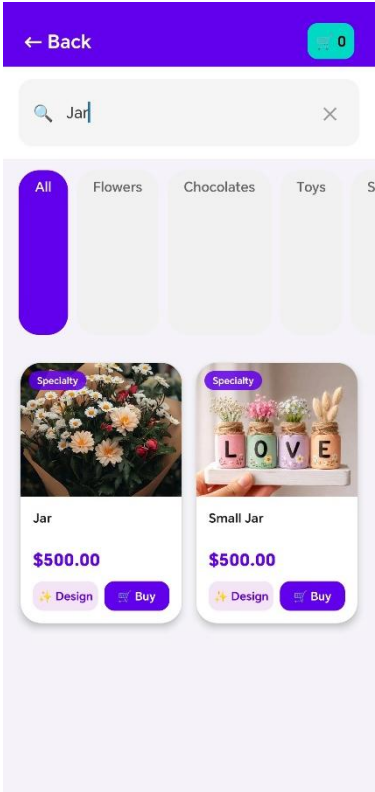
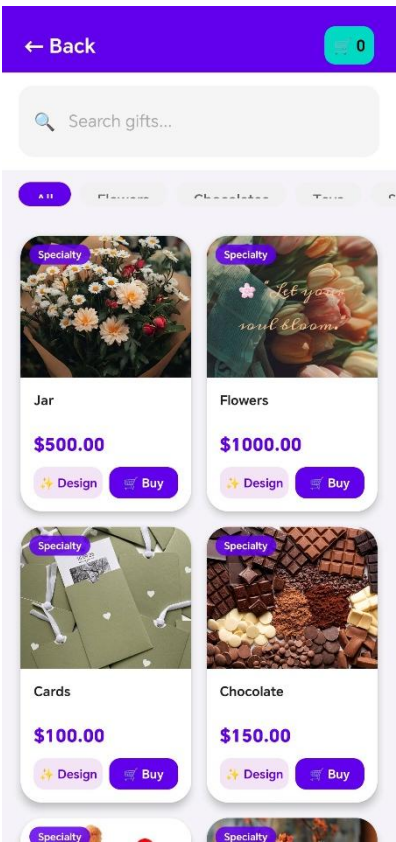
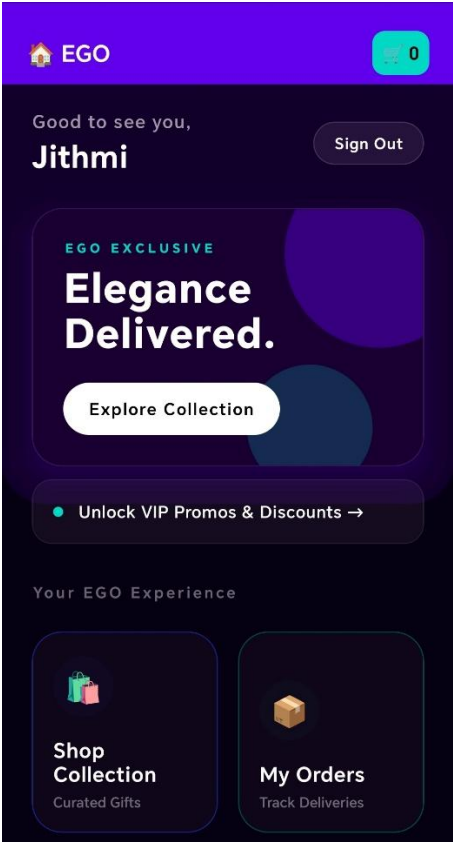
Expiry Days (e.g. 30)

Create Promo Code

Promo codes will be instantly available to users after creation



User Responsible pages



← Back

 Custom Item Request

Describe your dream gift and we'll make it happen!

Item Details

What item do you want?

Category

FlowersChocolatesToys

Describe your custom item in detail...

Budget Range

Under \$25\$25 ~ \$50\$50 ~ \$100\$100+

Reference Image (Optional)

← Back

 Custom Item Request

Describe your dream gift and we'll make it happen!

Reference Image (Optional)

 Tap to add reference

Personalization Details

Personal message or text you want...

Font Style

CursiveBoldElegantModern

Color Theme

GoldRoseTealNavyIvory

← Back

 Custom Item Request

Describe your dream gift and we'll make it happen!

Personal message or text you want...

Font Style

CursiveBoldElegantModern

Color Theme

GoldRoseTealNavyIvory

Urgency

NormalUrgentExpress

Submit Custom Request

← Back

 Promo Codes

Enter your code to unlock exclusive discounts

Your Promo Code

e.g. SUMMER

Check

How It Works

1 Enter your promo code above and tap "Check"

2 If valid, the discount will be shown

3 Enter the same code in Checkout to save!

Instant Discount

Secure Checkout

30 Day Validity

← Back

 Personalize Your Gift

Jar

\$500.00

Reference Photo

 Tap to add a photo

PNG, JPG accepted

Personal Message

Your message will appear here...

Write something heartfelt...

0/150

Font Style

CursiveBoldElegantModern

← Back

Your message will appear here...

Write something heartfelt...

0/150

Font Style

CursiveBoldElegantModern

Accent Color











Save Customization

Skip & Go to Checkout

← Back

Jar

Specialty

\$500.00

X

Subtotal

\$500.00

Total

\$500.00

Delivery Details

Street Address

City

Phone Number

Promo Code

Enter code

Apply

Payment Method

← Back

Jar

Specialty

\$500.00

X

Subtotal

\$500.00

Discount (NEW YEAR)

-\$50.00

Total

\$450.00

Delivery Details

Gampha

Gampha

0719853264

Promo Code

NEW YEAR" applied — 10% OFF!

Remove

← Back

Delivery Details

Gampha

Gampha

0719853264

Promo Code

NEW YEAR" applied — 10% OFF!

Remove

Payment Method

Cash on Delivery

Credit / Debit Card

Confirm Order — \$450.00

← Back

City

Phone Number

Promo Code

NEW YEAR" applied — 10% OFF!

Remove

Payment Method

Cash on Delivery

Credit / Debit Card

Card Number (16 digits)

MM/YY

CVV

Payments are securely verified

Pay \$450.00 securely